

The Citation Problem: Computing, the Internet, and the LLM Knowledge Crisis

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1. The Citation as a Computing Primitive

Before examining what has gone wrong with citations, it is worth understanding what a citation is in computational terms — because the framing reveals the problem with unusual clarity.

In computer science, a pointer is a variable that stores a memory address. It does not contain data; it contains a location where data can be found. A typed pointer carries additional information: not just where the data is, but what kind of data to expect there. A null pointer is a valid state — it explicitly represents the absence of a valid target.

A citation is a typed pointer in a knowledge system. It stores a location (the source document) and implicitly asserts a type (the relationship between the citing claim and the cited source). In the current citation system, that type is binary and unspecified: cited, or not cited. There is no type system. There is no null pointer. There is no way to represent "this address exists but the data at that address does not support the claim being made" or "no valid address exists for this claim."

This is a fundamental design flaw in the data structure of human knowledge. Every other mature computing system has moved beyond binary pointers to typed, graded, nullable references. The citation system has not. It is as if the entire scientific literature were written in a language with no type checking, no null safety, and no error handling — where every pointer is assumed to be valid and to contain exactly what the caller expects.

The graded citation is the type system that citations have always needed. It adds four states where there are currently two:

Citation Type	Computing Analogue	Epistemic Status
Verified	Strong typed reference — target contains expected data	Known known — claim directly supported
Contested	Weak reference with confidence interval	Known known with disputed confidence
Implied	Inferred reference — target implied by graph topology	Unknown known — evidence exists but not synthesised

Beyond evidence	Typed null — no valid target exists	Unknown unknown — honest boundary of knowledge
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The fourth type — the typed null — is the one that does not exist in the current system and that makes the entire structure unsafe. Without it, there is no way to distinguish between "no citation because the claim is so well established it needs none" and "no citation because no evidence exists." Both look identical. The result is that claims at the frontier of knowledge are indistinguishable from claims at its centre.

2. Is the Internet Cited Correctly?

The short answer is no, and the evidence is substantial.

The Internet is not a citation system. It is a hyperlink system. A hyperlink asserts only that one document is related to another — it carries no epistemic type, no confidence level, and no claim about the relationship between the content of the linking document and the content of the linked one. When web content cites scientific research, it typically does so through a chain of intermediaries, each of which introduces distortion.

The chain typically runs: primary paper → press release → science journalist → general news article → blog post → social media post → the text that ends up in a web corpus. Each step in this chain is a lossy transformation. Caveats are dropped. Effect sizes are inflated. Qualified findings become definitive claims. Specific populations become universal ones. The further from the primary source, the more distorted the representation.

This is not a new observation. Research on science journalism has documented it systematically. A 2022 study in the *Journal of Science Communication* found that the desire to create newsworthy stories about science led to a situation where poorer quality research — with more dramatic findings — received disproportionate coverage, while high-quality research with qualified or null results was systematically underreported. ¹ The Association of Science Communicators has described this as a "game of telephone" in which the original message is progressively distorted at each retelling. ²

The specific mechanisms of distortion are well documented. Press releases routinely overclaim preliminary studies. News articles frequently omit sample sizes, effect sizes, and limitations. Secondary sources cite the news article rather than the paper, inheriting its distortions. Tertiary sources cite the secondary source, compounding them further. By the time a claim reaches a general-audience website — the kind of content that dominates web corpora — it may bear only a loose relationship to what the original study actually found.

The scale of this problem on the Internet is difficult to quantify precisely, but the directional evidence is clear. A 2023 study on citation errors in scientific publications found that

citations in secondary sources are frequently incorrect because authors interpret the cited article through the lens of what they expect it to say rather than what it actually says. ³ The Office of Research Integrity has documented multiple patterns of citation misuse even within peer-reviewed literature — the problem is worse, not better, in general web content where no peer review exists at all. ⁴

The Internet also has a specific structural problem that the academic literature does not: **the absence of retraction infrastructure**. When a scientific paper is retracted, the retraction is noted in PubMed, Scopus, and Web of Science. When a blog post or news article is based on a retracted paper, nothing happens. The article remains live, continues to rank in search results, and continues to be cited by other web content. A 2012 study on the persistence of retracted articles on the Internet found that retracted papers continue to be referenced on academic websites, course pages, and general web content long after retraction, with no systematic mechanism for correction. ⁵

3. What LLMs Actually Trained On

Every major large language model has been trained on a substantial fraction of the publicly accessible Internet. The primary vehicle for this is Common Crawl — a non-profit organisation that has been crawling the web since 2008 and makes its data freely available. Common Crawl's dataset underpins the training of GPT, LLaMA, Falcon, Mistral, and most other frontier models. As of 2024, academic citations of Common Crawl had grown nearly 60-fold over a decade, reflecting its centrality to the AI training ecosystem. ⁶

What Common Crawl contains is a snapshot of the web as it exists — which means it contains the full accumulated distortion of the citation telephone game described above. It contains blog posts that misrepresent studies. It contains news articles that overclaim preliminary findings. It contains Wikipedia articles that cite secondary sources that cite tertiary sources that have drifted far from the original evidence. It contains the same distorted claim repeated across thousands of different websites, each one reinforcing the others.

A 2024 critical analysis of Common Crawl published at the ACM FAccT conference found that the dataset contains substantial quantities of low-quality, potentially harmful, and factually unreliable content, with no systematic quality filtering applied at the source level. ⁷ The Mozilla Foundation's analysis of Common Crawl as a training source noted that the dataset includes content that is "undesirable when training LLMs because it might lead to harmful outputs." ⁸

The citation problem is a specific instance of this broader data quality issue. When an LLM is trained on a corpus where the same distorted claim appears ten thousand times — because ten thousand blog posts all repeated the same misrepresentation of a single study — the

model learns the distorted version. It has no access to the original. It cannot know the difference. The distortion is baked into the weights.

4. The Publication Bias Compounding Effect

The problem runs deeper than the distortion introduced by science journalism and web content. The scientific literature itself — the primary source that the web content is distorting — is already systematically biased before any web distortion occurs.

Scientific publishing filters for success. Positive results dominate the published literature across disciplines. A 2012 analysis by Fanelli found that the proportion of papers reporting positive results exceeded 80% after 1999, peaking at 88.6% in 2005. ⁹ Roughly two-thirds of null results from NSF-funded experiments are never written up at all — they disappear into what Rosenthal called the "file drawer." ⁹ Chalmers and Glasziou estimated that 85% of total research investment is avoidably wasted, with failure to account for prior evidence — including unpublished negative results — identified as a primary driver. ⁹

A 2026 paper, "LLMs Have Made Failure Worth Publishing," formalises the consequence of this for AI systems:

"An LLM trained predominantly on positive-result literature inherits a distorted model of science in which most experiments succeed." ⁹

The same month, a companion paper titled "Dead Science Walking: Publication Bias and the AI Scientist Pipeline" quantified the distortion more precisely. The authors formalised what they call the **null result gap** — the difference between the actual rate of experimental success and the rate implied by the published literature — and estimated it across three domains: drug discovery (~0.60), psychology (~0.56), and cancer biology (~0.35). They then introduced an **amplification index** to model how retrieval, generation, and automated evaluation compound the raw gap, concluding that a standard three-stage AI pipeline can amplify corpus distortion by a factor of **2.18x**. ¹⁰

The governance failure modes they identify are directly relevant to the citation problem:

Confident rediscovery — an AI system confidently proposes a research direction that has already been tried and failed, because the failures were never published and therefore never learned.

Ghost evidence accumulation — an AI system cites evidence that appears to exist in the literature but is actually a distorted echo of a single weak finding, amplified through citation chains.

Replication laundering — an AI system treats a finding as replicated because it appears in multiple papers, without recognising that those papers all cite the same original source

rather than independently confirming it.

Confidence miscalibration — an AI system expresses high confidence in claims that the underlying evidence does not support, because it has been trained on a literature where high confidence is the norm regardless of evidence quality.

Each of these failure modes is a citation problem. Each one can be detected and corrected by a system that maintains a verified knowledge graph with explicit representation of evidence quality, citation chains, and the boundary of knowledge.

5. The Compounding Stack

The full picture is a compounding stack of distortions, each layer inheriting and amplifying the errors of the layer below.

Layer	Distortion mechanism	Cumulative effect
Primary research	Publication bias — 85%+ positive results, null results unpublished	Literature overrepresents success
Peer-reviewed citation	Amplification, bias, invention (Greenberg 2009)	Weak findings acquire unfounded authority
Press releases	Overclaiming, omission of caveats and limitations	Qualified findings become definitive claims
Science journalism	Telephone game — each retelling adds noise, removes nuance	Effect sizes inflated, populations generalised
Web content	No retraction infrastructure, no quality filtering	Distorted claims persist and multiply
LLM training corpus	Common Crawl ingests all layers above	Model learns the distorted version as ground truth
LLM output	Confident synthesis of distorted training data	Confident, fluent, wrong

The LLM is at the bottom of this stack. It has consumed all the layers above it. When it generates a confident claim about protein synthesis, drug efficacy, or legal precedent, it is synthesising the accumulated distortions of every layer in the stack. It does not know this. It cannot know this. It has no access to the primary evidence. It has no map of the citation chains. It has no record of what was never published.

This is not a failure of the LLM. It is a failure of the knowledge infrastructure the LLM was built on.

6. The Corrective Layer

What the stack above is missing — at every level — is a verified knowledge graph anchored to primary sources with intact, graded citation chains. This is precisely what `truthdesk.claims` is building.

The platform ingests primary sources directly: PubMed full-text papers, PDB protein structures, UniProt protein annotations, OpenFDA drug data, ClinicalTrials.gov trial records. It does not ingest secondary sources, press releases, or web content. It builds its knowledge graph from the primary evidence and maintains explicit records of what each source says, with what confidence, and in what context.

The verdict engine produces graded verdicts — Supported, Contradicted, Partially Supported, Insufficient Evidence — which map directly onto the four citation types described in Section 1. The Frontier Engine identifies unknown knowns: claims that are implied by the knowledge graph but have not been directly tested. The Dream State Engine probes unknown unknowns: the questions that the current evidence cannot answer.

The result is a knowledge layer that sits above the distortion stack rather than within it. An LLM that queries this layer before generating a response gets grounded, verified, typed answers — not the distorted synthesis of ten thousand secondary sources. An LLM that does not query it is flying blind, generating confident claims from a training corpus that has been distorted at every level from primary research to web content.

The strategic position is therefore not just a citation checker or a fact-verification tool. It is the corrective infrastructure for the entire LLM knowledge stack — the typed, graded, null-safe pointer system that the current citation architecture has always lacked and that AI systems urgently need.

7. The Computing Connection

The connection to computing goes beyond the pointer analogy. The problem of knowledge representation under uncertainty is a classical problem in computer science, addressed by type theory, probabilistic programming, knowledge graphs, and formal verification systems. Each of these disciplines has developed rigorous tools for representing what is known, what is uncertain, and what is absent — tools that the citation system has never adopted.

Type theory distinguishes between values of different kinds and enforces that operations are applied only to values of the appropriate type. A typed citation system would distinguish between a citation that directly supports a claim, one that provides background context, one that is contested, and one that marks the boundary of knowledge. Operations on these types — such as aggregating evidence or computing confidence — would be type-safe: you could not treat a contested citation as if it were a verified one without an explicit type conversion.

Probabilistic programming represents uncertainty as a first-class value. A probabilistic citation would carry not just a verdict but a probability distribution over verdicts, updated as new evidence arrives. This is closer to what the `truthdesk.claims` confidence scoring system already does — each claim has a confidence score that is recalibrated as the knowledge graph grows.

Knowledge graphs represent entities and relationships explicitly, enabling inference over the structure of knowledge rather than just retrieval of individual facts. The `truthdesk.claims` knowledge graph is the infrastructure needed to surface unknown knowns — the claims that are implied by the graph topology but have not been directly tested.

Formal verification systems prove properties of programs by checking them against a specification. A citation integrity system is, in essence, a formal verifier for knowledge claims — it checks whether a claim is consistent with the specification provided by the primary evidence.

The graded citation is not a new idea in computing. It is the application of ideas that are standard in computing — type safety, probabilistic reasoning, graph inference, formal verification — to a domain that has been operating without them for four centuries.

8. Conclusion

The Internet is not cited correctly. It never was. The web corpus that LLMs trained on is a compounding stack of distortions: publication bias in the primary literature, citation amplification and invention in the academic record, overclaiming in press releases, telephone-game degradation in science journalism, and zero quality filtering in web content. LLMs have consumed all of this and learned the distorted version as ground truth.

The null result gap — the difference between what the evidence actually shows and what the published and web literature implies — is estimated at 0.35 to 0.60 across major scientific domains. A standard AI pipeline amplifies this distortion by a factor of 2.18x. The result is AI systems that are confidently wrong in ways that are systematic, predictable, and invisible to any tool that only checks whether citations exist.

The corrective layer is a verified knowledge graph anchored to primary sources, with graded citations that carry explicit epistemic type information — including the typed null that represents the honest boundary of knowledge. This is not a marginal improvement on existing citation tools. It is a different category of infrastructure: the type system that the knowledge stack has always needed and that AI systems make urgently necessary.

The unknown unknown, properly cited, is not an admission of failure. It is the most honest and most useful thing a knowledge system can say.

References

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